Machine Learning for Wind Farm Output Forecasting
Executive Summary

42578 Advanced Business Analytics

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Importance of the problem

Nowadays most countries, especially in Europe, are taking concrete steps to reduce CO2 and stop global warming. As it is usually the case, large corporations have the biggest influence as they are the largest polluters, therefore it makes sense to target them to reduce emissions. One of the key steps that the governments are taking is the CO2 tax, for example, in Denmark the government will introduce a corporate carbon tax from 2025 with as much as 1125 DKK/tCO2 (150 €/tCO2). While this measure is necessary for the well-being of our planet, it also poses a challenge for affected companies and creates a business disadvantage for them when compared to countries without similar policies.

Still, our team believes that there is a solution. What if there is a legally accepted process that enables companies to accurately predict the best time to conduct their energy-intensive operations to minimize their carbon footprint and avoid paying the CO2 tax altogether?

Main findings

We have developed machine learning models that can predict weather patterns and estimate the CO2-neutral energy production from wind farms. This cutting-edge technology enables companies to conduct their operations in a carbon-neutral way and significantly reduce, or even eliminate, their CO2 tax liability. We are able to predict the next 4 hours of wind speed conditions with relatively good precision, which enables us to feed the data into our capacity prediction models, providing an estimate of the expected wind farm energy output. Based on large amounts of weather data including historical wind speed patterns, we are able to generate accurate predictions that can be used to optimize energy usage and minimize environmental impact.

Moreover, by using the data from multiple wind farms allocated near a specific company or business, and connected through the grid, in the future, we can easily predict the total production of these wind farms in a given time frame.

Furthermore, we developed a procedure which finds new possible locations at which we recommend to build new wind farms. We did that by filtering possible locations based on different topographical and infrastructural criteria and applying our models to those, we could obtain their predicted average production.

Recommendations

For our clients, we can give the exact time when they should schedule their operations. This data can be used as a legal document to prove that the company is meeting its CO2 neutrality goals. For example, if you are a large AI company training AI algorithms, we could give an exact time frame for when they should schedule their training. Or, if you are a hotel or a restaurant that needs to power on their washers, our models are equally helpful. This enables our clients to optimize their energy usage and minimize carbon emissions while still meeting business objectives.

In addition to that, we can also make recommendations to wind power companies as to where the should consider to build new wind farms in order to get the maximum possible energy output. This can both increase the revenue of the wind power company itself, and also help companies situated around these locations to meet their sustainability goals.